

CWIIIF: A Novel Algorithm for Identifying Influential Nodes in Multilayer Networks

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Abstract—The identification of influential nodes in multilayer networks is a rapidly growing area in network science. However, insufficient consideration of both inter- and intra-layer weights in existing research has limited the effectiveness of node identification methods. To address this gap, we propose a novel algorithm, coupling weighted intra-layer and inter-layer influence factors (CWIIIF), which accurately identifies nodes that exert significant influence in multilayer networks. The algorithm integrates weighted intra- and inter-layer influence factors, taking into account the unique properties of multilayer network structures. First, we define a set of layer weight influence parameters, including active nodes, active paths, and communication intersections between layers, to determine the weight of each network layer. We then calculate the intra-layer influence of each node using a combination of K-shell and betweenness centrality methods. Finally, we introduce a set of coupled equations that convert the intra-layer influence vectors into scalar values by incorporating the weights of each layer, producing a final influence score for each node. To validate the effectiveness of our algorithm, we conducted four comparative experiments across nine real-world and one synthetic multilayer networks. The results demonstrate that our algorithm significantly outperforms nine classical and state-of-the-art methods for identifying influential nodes.

Index Terms—Coupled weighting, influential nodes, inter-layer influence factors, intra-layer influence factors, multilayer network.

I. INTRODUCTION

OVER the past decade, research into complex networks has experienced a significant surge. The challenge of identifying node importance has become a focal point within this field. This entails recognizing influential nodes capable of affecting a vast array of others essential for grasping the spreading dynamics within complex networks [1], optimizing seed selection for influence maximization [2], [3], among other applications. In essence, the issue revolves around pinpointing specific nodes that, when activated, can efficiently transmit information ranging from rumors [4] to advertisements [5], products [6], and viruses [7], [8] across a substantial segment of the network. Conversely, isolating these nodes can effectively mitigate the spread of unwanted agents, such as viruses [9], [10]. Consequently, these nodes hold particular significance in contexts involving the spread of information [11], control of rumors [12], management of epidemics [13], and viral marketing strategies [14].

Traditionally, research on complex networks has centered on single-layer models, where all nodes (entities) and their links (communication channels) are assumed to reside within a singular network. Along this research trajectory, many advanced methods worth learning, such as the neural diffusion model [15] and convolutional neural networks [16], have been used to identify influential nodes, achieving promising results. However, recent developments have seen a shift toward the exploration of novel complex network models, wherein nodes are considered to span multiple networks or layers. These networks are characterized as multiplex [17], multisliced [18], multilevel [19], interdependent [20], or more broadly multilayer networks [21], [22]. Examples of such networks include online social platforms, financial systems, and transportation networks [23], [24]. A multilayer network presents a sophisticated framework where each node, representing an entity, engages in various forms of interactions, each depicted as a separate layer. For instance, in transportation networks, nodes may represent stations with layers corresponding to different transport modes such as subways, buses, or trains. This model allows for an in-depth analysis of how different interaction layers influence the overall network dynamics, providing critical insights into

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multifaceted interactions across various fields such as sociology, biology, computer science, and physics. These insights facilitate understanding phenomena ranging from the spread of infectious diseases through social contacts to the propagation of information in social networks and behaviors in financial markets.

A. Related Works

Various approaches have been proposed to address the issue of identifying significant nodes within multilayer networks.

One category of methods extends existing single-layer network techniques and concepts to multilayer contexts [25], [26]. Methods like the one proposed by Chakraborty and Narayanam [27] have evolved centrality metrics such as betweenness centrality for multilayer networks. These methods utilize a parameter to balance the importance between intra-layer and inter-layer edges. However, they require measuring node observables along intra-layer paths and their values across inter-layer traversals. Solá et al. [28] developed eigenvector centrality for multilayer networks, where centrality calculations are based on a vector approach for each layer combined nonuniformly with each layer's diffusion coefficients. In another advancement, Li et al. [29] introduced a multilayer network's PageRank algorithm, modified by a gravity model to adapt the PageRank approach. This improved centrality measure incorporates the gravity model to identify nodes within multilayer networks that have physical or logical spatial interactions. De et al. [30] presented a versatile algorithm that extends single-layer eigenvector and PageRank centralities to multilayer networks. They also proposed a mathematical framework allowing the calculation of centrality within such networks, which ranks nodes to identify those that play a pivotal role in the cohesion of the entire structure by linking different types of relationships. Basaras et al. [31] adapted the power community index, a locality-specific centrality metric, to multilayer networks. Bergermann and Stoll [32] utilized the supra-adjacency matrix as a novel representation to extend eigenvector centrality to multilayer networks. This method allows the extension of matrix function-based centralities to multilayer networks by assigning appropriate edge weights. Despite their ability to identify key nodes in multilayer networks, these methods can overlook the networks' inter-layer structures, leading to inaccuracies.

The second category of methods integrates novel concepts to evaluate node significance in multilayer networks, such as center vector compression, layer integration, result aggregation, and evidence theory. For instance, Zhou et al. [33] developed a technique to map and compress centrality vector values into scalar values, enabling efficient centrality calculation and node ranking across multiple layers. This approach determines factors influencing each network layer based on specific layer characteristics, subsequently compressing the centrality vector values of each node into scalar values using the identified layer influence factors. Lv et al. [34] introduced a centrality computation method that employs random walks and fourth-order tensors. This method starts by calculating the hubs and keypoints for each node within a layer and then combines

these scores to pinpoint significant nodes. Domenico et al. [35], [36], [37] explored layer aggregation methods for centrality calculation; for example, in a related study [35], the approach visualizes connections between nodes across different layers as a simplex network, allowing the application of single-layer network centrality algorithms to the aggregated network. Li et al. [38] proposed viewing the impact of each node across different layers as types of information or as information from different timelines, positing that the cumulative impact of a node on the entire multilayer network is equivalent to the fusion of these disparate single-layer effects. To achieve this, they employed Dempster-Shafer evidence theory to merge the influences from different layers into a unified node importance ranking. However, due to specific conditions and requirements related to network structure, this category of methods may not be universally applicable and may exhibit practical limitations.

The third category of methods encompasses algorithms designed to compute the centrality of active nodes and spreaders across all layers of a multilayer network [39], [40]. For instance, Zhao et al. [40] introduced an algorithm to rank the nodes within multilayer networks, identifying the minimal requisite set of influential nodes. This approach is also applicable to other centrality metrics, such as degree and closeness. Wan et al. [41] developed a method to determine node significance in multilayer heterogeneous networks by integrating multiple types of relational data. They established a multilayer heterogeneous network model and framework using tensor algebra, computed centrality for each layer's nodes, quantified the weights of neighboring nodes across layers, and synthesized multiple relational data types to enhance node attributes and transmission dynamics, thereby deriving a ranking of node importance. Based on multitask optimization theory, Wang and Tan [42] proposed the multifactor evolutionary (MFEA-RIM_m) algorithm, which is capable of parallelly optimizing multiple scenarios and effectively addressing the challenge of seed selection in multilayer networks. In another study [43], a random walk-based method was proposed to pinpoint key nodes in multilayer networks through the analysis of effective interlayer parameters in propagators, illustrating the complexity of computing centrality in such networks. This research underscores that centrality measures appropriate for single layers do not necessarily translate to multilayer contexts, necessitating a comprehensive evaluation of node centrality and relevance across all layers. Rani and Kumar [44] discovered that social media networks are interconnected via heterogeneous interactions among nodes, which furnish critical contextual insights about these nodes. Consequently, they amalgamated various interaction types and employed multilayer networks to explore the diverse characteristics of social networks. In a related study [45], Wang et al. introduced a Memetic algorithm and devised an augmented independent cascade model tailored for multilayer networks. This model aims to assess the extensive influence of seed nodes across different network layers. During this model's approximation process, several real-world variables were taken into account, including overall network activity levels, variable propagation probabilities, and the weighting of network layers, thereby enhancing the model's applicability in practical

scenarios. By evaluating the influence ability of seed nodes under structural damage, a Memetic algorithm named MFEA-RIM_{multi} [46] is proposed, which successfully addresses the problem of robust influence maximization in multilayer networks. Although this category of methods considers inter-layer edges, it treats them as ordinary edges, which may fail to fully capture the unique nature of inter-layer interactions. Furthermore, it may have limitations in capturing the global characteristics of multilayer networks.

B. Motivation and Contributions

In this article, we investigate a multilayer network with identical nodes across layers but distinct edges, resulting in nodes with similar properties bearing different labels. Consequently, we develop a technique to assign varying weights to network layers based on specific criteria. These weights embody the significance and influence of each layer on the network's structure and dynamics, enabling hierarchical weighting to effectively identify crucial nodes in multilayer networks. Nonetheless, hierarchical weighting approaches face challenges, such as handling information loss, devising suitable weighting schemes, and integrating weighting results from disparate layers. Currently, one method aggregates different layers, which, although capable of identifying important nodes, may overlook information due to incomplete data. Another approach uniformly weights layers, assigning equal significance to all layers and ensuring their equal contribution to the entire network. However, given the network structure's characteristics, layers might possess differing weights or influences. Some methods assign varying weights to different layers, but only consider specific structural properties of the network, such as node aggregation degree.

To address these challenges, this study addresses the viability of developing an efficient algorithm that couples weighted factors within and between layers to identify influential nodes in multilayer networks. The key contributions of this article are outlined as follows.

- 1) We introduce a novel collection of intra-layer and inter-layer influence parameters, integrating these parameters to formulate layer weighting coefficients, thereby assigning appropriate weights to each layer in a reasonable manner.
- 2) We construct a series of coupled equations that incorporate layer weight coefficients to convert the intra-layer influence vectors of nodes into scalar values, enabling us to rank the nodes based on their influence capability.
- 3) We present a new algorithm, known as coupling weighted intra-layer and inter-layer influence factors (CWIIIF). This algorithm effectively couples weighted factors within and between layers in multilayer networks, proficiently identifying the influential nodes.
- 4) Through numerous experiments conducted on nine real-world and one synthetic multilayer network datasets, we consistently demonstrate that our algorithm, CWIIIF, excels in identifying influential nodes, outperforming nine

classical and state-of-the-art methods regardless of the size or characteristics of the multilayer networks studied. These findings underscore the effectiveness of our algorithm.

The remainder of this article is structured as follows. Section II delves into the details of the proposed algorithm. Section III presents the experimental results and evaluations of our algorithm, juxtaposed with other existing methods. Section IV offers a summary of the entire text and outlines future research prospects.

II. PROPOSED ALGORITHM

Consider a multilayer network, $G = (L, V, E)$, where L represents the number of layers, and V is the set of nodes, with each node as a replicated entity mapped across all layers. Thus, $V = V^{[1]} = V^{[2]} = \dots = V^{[L]}$. The term E denotes the set of edges within all layers, which may vary between layers; therefore, $E = \{E^{[1]}, E^{[2]}, \dots, E^{[L]}\}$. Given the one-to-one correspondence of all nodes across layers, we define $E_i^{[\alpha, \beta]}$ as the inter-layer edge between layer α and layer β corresponding to node i in a multilayer network, where $\alpha, \beta \in \{1, 2, \dots, L\}$ and $\alpha \neq \beta$. It is worth noting that the inter-layer edge here does not refer to an edge from node i to itself, but rather describes the connection of node i across different layers. Node influence in multilayer networks is related to both intra-layer and inter-layer structures. However, current studies lack effective methods or algorithms to fully quantify this influence and thus accurately identify influential nodes. To address this gap, we propose the CWIIIF algorithm, which consists of three key steps. 1) Calculating the layer-weighted coefficients for each layer. 2) Determining the intra-layer influence vector for each node. 3) Integrating and mapping node influence values. This section provides a comprehensive explanation of each step, detailing the methodologies and techniques involved. The framework diagram of the CWIIIF algorithm in a three-layer network is presented in Fig. 1.

A. CWIIIF Algorithm

Step 1: Calculation of Layer-Weighted Coefficients for Each Layer

In this step, the layer-weighted coefficients are calculated by considering the factors that influence each layer, with the goal of improving the accuracy of identifying key nodes. Layers with a greater number of active nodes can disseminate information more rapidly and efficiently, leading to higher transmission rates. To evaluate the influence of each layer, not only is the number of active nodes considered, but also the length of active paths and interlayer communication. The key factors influencing this process are detailed below:

1) *Active Nodes*: Extensive research indicates that nodes do not exhibit uniform connectivity across all layers of a multilayer network. For instance, in a three-layer network composed of bus stations, subway stations, and train stations, a node might show high connectivity in the train station layer while exhibiting

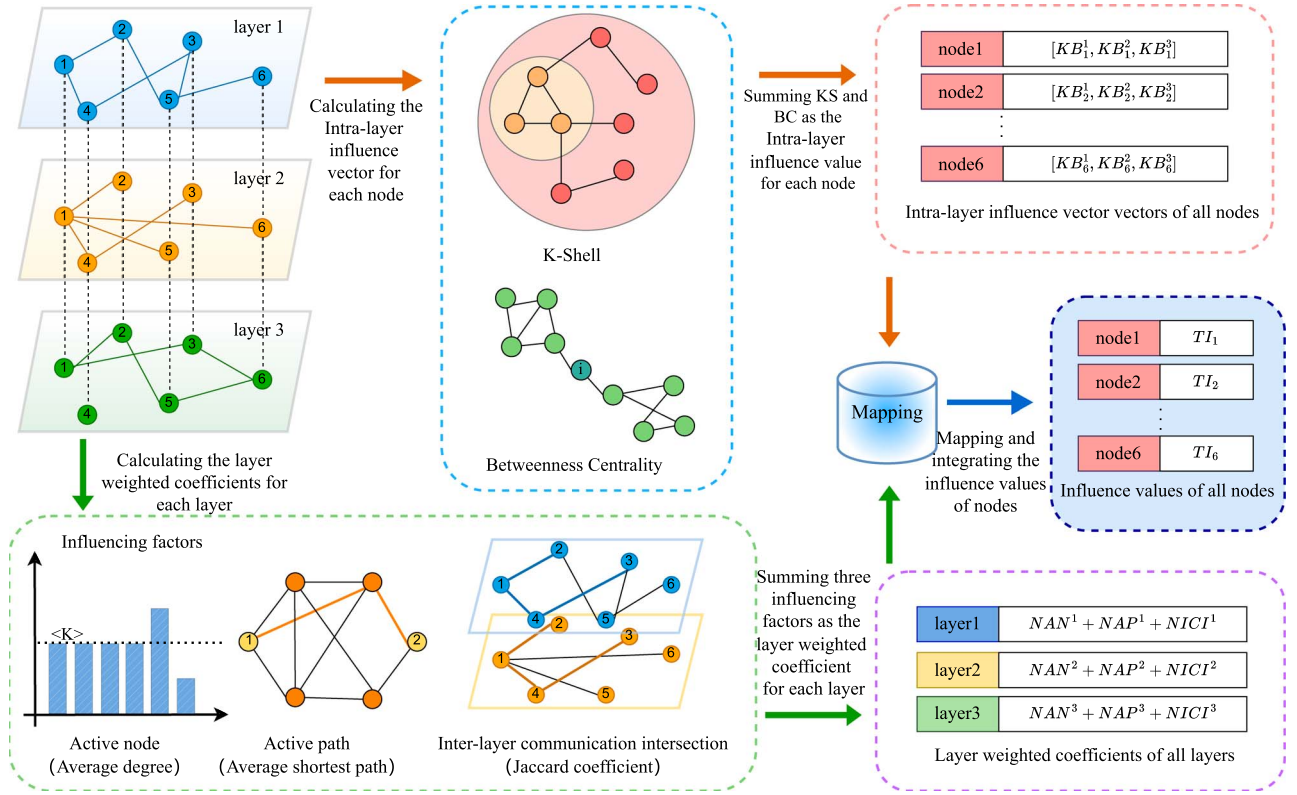


Fig. 1. Framework diagram of CWIIF algorithm under a three-layer network.

lower connectivity in the other two layers. As a result, layers with a greater number of active nodes tend to spread information faster, potentially affecting more nodes in a shorter time.

Definition 1 (Active Node): Node i in layer α is defined as active if it connects to a greater number of nodes within that layer than the average degree of all nodes in the layer. Formally, this condition is expressed as $K_i^{[\alpha]} > \langle K^{[\alpha]} \rangle$, where $K_i^{[\alpha]}$ denotes the degree of node i in layer α , and $\langle K^{[\alpha]} \rangle$ represents the average degree of layer α .

The activity vector for each node can be represented as follows:

$$b_i = \{b_i^{[1]}, b_i^{[2]}, \dots, b_i^{[L]}\} \quad (1)$$

where $b_i^{[\alpha]}$ denotes whether node i is in an active state in layer α , and $\alpha \in \{1, 2, \dots, L\}$. $b_i^{[\alpha]}$ can be represented as follows:

$$b_i^{[\alpha]} = \begin{cases} 1, & \text{node } i \text{ is active in layer } \alpha \\ 0, & \text{node } i \text{ is inactive in layer } \alpha. \end{cases} \quad (2)$$

Let $N = |V|$ denote the total number of nodes in the network, including isolated nodes. Therefore, the number of active nodes in each layer of the network is represented as

$$b^{[\alpha]} = \sum_{i=1}^N b_i^{[\alpha]}. \quad (3)$$

It is necessary to normalize the aforementioned parameter. Normalization entails scaling the results of each individual

layer to values between 0 and 1, thereby ensuring consistency and comparability of influence factors across different layers. According to (3), the normalization of the number of active nodes per layer is denoted by the normalized active node (NAN) label, expressed as

$$NAN^{[\alpha]} = \frac{b^{[\alpha]}}{\sum_{\beta=1}^L b^{[\beta]}}. \quad (4)$$

Here, $b^{[\alpha]}$ denotes the number of active nodes in layer α , and $\sum_{\beta=1}^L b^{[\beta]}$ represents the total number of active nodes across all L layers. The $NAN^{[\alpha]}$ score is within the range of (0, 1), where higher values indicate a greater quantity of active nodes and correspondingly increased influence factors.

2) Active Path: In each layer of information propagation, shorter transmission paths not only accelerate the speed of transmission but also enhance coverage and effectiveness. Therefore, the quality of the transmission path is a critical factor to consider. By incorporating the length of active paths as a layer-weighted influence factor, the efficiency of information dissemination across the entire network can be more accurately assessed.

Definition 2 (Active Path): If both nodes i and j are not isolated, there exists a shortest path between them within layer α , which is defined as an active path. The length of this active path is denoted as $d_{ij}^{[\alpha]}$. If either node i or node j is isolated, no shortest path exists between them, and thus, there is no active path.

The average active path length for each layer is calculated as follows:

$$d^{[\alpha]} = \frac{\sum_{i,j=1 \& j \neq i}^N d_{ij}^{[\alpha]}}{2N} \quad (5)$$

where $\sum_{j=1, j \neq i}^N d_{ij}^{[\alpha]}$ represents the total length of shortest paths between all pairs of nodes in layer α . Notably, isolated nodes within a layer will be excluded from the calculation of the average shortest path length. According to (5), the normalization of the average active path length per layer is denoted by NAP, which can be computed as follows:

$$\text{NAP}^{[\alpha]} = \frac{d^{[\alpha]}}{\sum_{\beta=1}^L d^{[\beta]}} \quad (6)$$

where $d^{[\alpha]}$ denotes the average active path length in layer α , and $\sum_{\beta=1}^L d^{[\beta]}$ represents the total average active path length across all L layers.

3) *Inter-layer Communication Intersection*: In a multilayer network, each layer may share edges with other layers. For instance, nodes in the bus station layer may be interconnected, and similar connections may exist in layers representing subway stations, train stations, or others. Therefore, overlapping edges represent the intersection of intralayer communication with other layers, directly affecting the inter-layer penetration coefficient of the network.

Definition 3 (Inter-layer Communication Intersection, ICI): In our proposed metric, we employ the Jaccard coefficient [47] to quantify this communication intersection relative to other network layers. Given the potential for numerous overlapping edges between each layer and others, this can result in elevated influence values. The inter-layer communication intersection between layer α and the other $L - 1$ layers can be computed as follows:

$$\text{Jaccard}^{[\alpha]} = \sum_{\beta=1 \& \beta \neq \alpha}^L \frac{E^{[\alpha]} \cap E^{[\beta]}}{E^{[\alpha]} \cup E^{[\beta]}} \quad (7)$$

where $E^{[\alpha]}$ and $E^{[\beta]}$ represent the edge sets of layers α and β , respectively. By normalizing the inter-layer communication intersection between layers, the normalized ICI values for each layer are represented using NICI as follows:

$$\text{NICI}^{[\alpha]} = \frac{\text{Jaccard}^{[\alpha]}}{\sum_{\beta=1}^L \text{Jaccard}^{[\beta]}} \quad (8)$$

where $\text{Jaccard}^{[\alpha]}$ denotes the Jaccard coefficient in layer α , and $\sum_{\beta=1}^L \text{Jaccard}^{[\beta]}$ represents the total Jaccard coefficient across all L layers.

To ensure fairness among the different factors, normalization is applied to the three influencing factors of layer weights mentioned above. This prevents the impact of differing dimensions from skewing the weight calculation. The weight coefficient $W(\alpha)$ for layer α can be calculated as follows:

$$W(\alpha) = \text{NAN}^{[\alpha]} + \text{NAP}^{[\alpha]} + \text{NICI}^{[\alpha]}. \quad (9)$$

Since the three influencing factors evaluate distinct dimensional characteristics within the multilayer network, and given the

Algorithm 1: LWC.

Input: Network $G = (L, V, E)$;

Output: $W(\alpha)$;

Initialization;

for each layer $\alpha \in \{1, 2, \dots, L\}$ **do**

for each node $v_i^{[\alpha]} \in V^{[\alpha]}$ **do**

 Compute the number of active nodes ($b^{[\alpha]}$);

if $K_i^{[\alpha]} > \langle K^{[\alpha]} \rangle$ **then**

$b^{[\alpha]} = b^{[\alpha]} + 1$;

end if

 Compute the average active path length ($d^{[\alpha]}$);

 Compute the Jaccard coefficient ($\text{Jaccard}^{[\alpha]}$);

end for

 Compute the normalized values ($\text{NAN}^{[\alpha]}$, $\text{NAP}^{[\alpha]}$, $\text{NICI}^{[\alpha]}$);

 Compute the layer weighted coefficient for each layer ($W(\alpha)$);

end for

absence of significant differences in their importance, they are assigned equal weights. This allocation prevents any single factor from dominating the final weight outcome, thereby avoiding undue amplification or attenuation of specific influences and ensuring a balanced consideration of all dimensions. Algorithm 1 presents the pseudocode for the layer weighted coefficient (LWC) algorithm.

Step 2: Calculating the Intra-layer Influence Vector for Each Node

During this stage, it is necessary to independently compute the intra-layer influence values for each node on each layer, without considering inter-layer connections. The K-shell method possesses characteristics of both network centrality and local significance; nodes with higher degrees and more neighbors are considered more important. The betweenness centrality method can compensate for the K-shell method's neglect of critical network topologies based on global properties. Given the negligible impact of betweenness centrality on K-shell values in large networks, independently normalizing the K-shell and betweenness centrality enhances balance [48], [49]. This normalization strategy ensures a logical integration, accurately reflecting the nodes' pervasive influence across layers. Considering the combination of K-shell and Betweenness Centrality to measure node influence helps capture both the local structural centrality and the global path contribution of nodes, compensating for the shortcomings of using a single metric. Therefore, we combine the K-shell method with the betweenness centrality method to uniquely determine the influence $KB_i^{[\alpha]}$ of node i on layer α , which can be computed as follows:

$$KB_i^{[\alpha]} = KS_i^{[\alpha]} + BC_i^{[\alpha]}. \quad (10)$$

Here, $KS_i^{[\alpha]}$ represents the normalized K-shell value of node i in layer α , where the normalization is performed by the sum of K-shell values of all nodes in that layer. Similarly, $BC_i^{[\alpha]}$ denotes the normalized betweenness centrality of node i in layer α , normalized by the sum of betweenness centrality values of

Algorithm 2: *CWIIF*.

Input: Network $G = (L, V, E)$;
Output: TI_i ;
Initialization;
Compute $W(\alpha)$ according to Alg. 1;
for each layer $\alpha \in \{1, 2, \dots, L\}$ **do**
 for each node $v_i^{[\alpha]} \in V^{[\alpha]}$ **do**
 Compute the intra-layer influence vector for each node ($KB_i^{[\alpha]}$);
 Map and integrate the influence values of nodes (TI_i);
 end for
end for

all nodes in that layer. Therefore, based on the number of layers in a multilayer network, the intra-layer influence vector $KB_i^{[\alpha]}$ of node i can be expressed as follows:

$$KB_i = [KB_i^{[1]}, KB_i^{[2]}, \dots, KB_i^{[\alpha]}, \dots, KB_i^{[L]}] \quad (11)$$

where $KB_i^{[\alpha]}$ represents the influence value of node i in layer α . It is worth noting that we set K-shell and betweenness centrality to equal weights, avoiding the interference of arbitrary weight biases.

Step 3: Mapping and Integrating the Influence Values of Nodes

In this step, it is necessary to compute the final influence value of the nodes. Due to the heterogeneity inherent in multilayer networks, different layers often display marked disparities, resulting in variable node influence across layers. Consequently, it is crucial to merge the layer weight coefficients, established in Step 1, to consolidate the within-layer influence vectors of each node into a single scalar value. To achieve this, we define TI_i to calculate the total influence of node i in the entire network, which can be expressed as follows:

$$TI_i = \sum_{\alpha=1}^L KB_i^{[\alpha]} \times W(\alpha). \quad (12)$$

For clarity, Algorithm 2 displays the pseudocode of the CWIIF algorithm.

B. Algorithm Complexity Analysis

The proposed CWIIF algorithm mainly consists of three steps. In the first step, the weights of each layer of the network are calculated. NAN represents the average network complexity, with a time complexity of $O(L * N)$, where L is the number of layers in the network and $N = |V|$ is the total number of nodes in each layer. NAP requires calculating the shortest path length between all pairs of nodes, with a time complexity of $O(L * N^3)$ by using Floyd–Marshall algorithm. NICI involves calculating the Jaccard coefficient, which requires computing the size of the intersection and union of two sets. The time complexity is articulated as $O(L^2 * (|E^{[\alpha]}| + |E^{[\beta]}|))$, where $|E^{[\alpha]}|$ and $|E^{[\beta]}|$ represent the number of edges in layer α and layer β , respectively. This can be further simplified as $O(L^2 * |E|)$. Here, $|E|$ represents the average number of edges in all layers

TABLE I
BASIC PROPERTIES OF NINE REAL-WORLD MULTILAYER NETWORKS, WHERE $|E|$ DENOTES THE TOTAL NUMBER OF EDGES IN THE NETWORK

Network	N	$ E $	L	Nature
Vickers-Chan-7thGraders_multiplex	29	740	3	Social
Kapferer-Tailor-Shop_multiplex	39	1018	4	Social
Lazega-Law-Firm_multiplex	71	2571	3	Social
CS-Aarhus_multiplex	61	620	5	Social
London_multiplex_Transport	369	441	3	Transport
PierreAuger_multiplex_Coauthorship	514	7153	16	Coauthorship
Drosophila_genetic_multiplex	1364	3059	2	Genetic
Sacchpomb_genetic_multiplex	875	8363	3	Genetic
Homo_Genetic_multiplex	3859	23 634	3	Genetic

of the network. The time complexity for calculating the weight W is $O(L)$. In the second step, the intra-layer influence vector of each node is calculated. The K-shell method removes the node with the smallest degree iteratively, with a time complexity of $O(L * (N + |E|))$, and the betweenness centrality is also computed to calculate the shortest path time complexity of $O(L * N^3)$. In the third step, the intra-layer influence vectors of each node are mapped to scalar values with a vector length equal to the number of network layers, with a time complexity of $O(L * N)$. Since L is typically small in complex networks, the time complexity of the CWIIF algorithm can be simplified to $O(N^3 + |E|)$. Considering that the complexity of our algorithm is not superior to that of the comparison algorithms, we did not include a runtime comparison in the experiments section.

III. EXPERIMENTS

To evaluate the effectiveness of the proposed CWIIF algorithm, we conducted experiments on nine real-world and one synthetic multilayer networks using several evaluation metrics. These metrics—Spearman’s rank correlation coefficient, Kendall correlation coefficient, imprecision function, monotonicity index, and network connectivity—measure the algorithm’s performance in identifying critical nodes. Moreover, we compare the CWIIF algorithm against nine classical and state-of-the-art methods to validate its effectiveness and demonstrate its superiority.

A. Experimental Setup

1) *Datasets*: Our algorithm can be applied to various real-world multilayer network systems. In this article, we selected nine real-world datasets for experimentation. The Vickers, Kapferer, Lazega, and CS networks are social networks, where high-influence individuals are used to study group behavior and diffusion mechanisms. The London network is a transportation network, with key nodes used for traffic flow analysis and system optimization. The PierreAuger network is an academic author collaboration network, where core authors help promote research collaboration and resource optimization. The Drosophila, Sacchpomb, and Homo networks are biological networks, where key genes provide important targets for disease mechanism research and drug development. Table I summarizes the primary characteristics of the nine real-world multilayer

networks used in the experiments. Further details are available at: <https://manliodomenico.com/data.php>. Note that the seventh to ninth networks are subsets extracted from the original network.

It is important to note that in practical systems, the following steps can be used to apply our algorithm.

- 1) *Data Collection and Modeling*: For the target system, collect data from each layer and transform it into a multilayer network format.
- 2) *Constructing the Multilayer Network*: Organize the data from each layer into a multilayer network, where each layer represents a different type of interaction or relationship within the network.
- 3) *Applying the Algorithm for Analysis*: Use our algorithm to assess the importance of nodes in the multilayer network. For each node, our algorithm provides a comprehensive measure of its overall influence within the network.

For the sake of reproducibility, all the code (including all models, methods, etc.) and data used in the article are available from the following website: <https://github.com/HY-47/cwiiif.git>.

2) *Benchmark Methods for Comparison*: In our experiments, we employ nine classical and state-of-the-art methods for comparison with the CWIIF algorithm: sumPR, sumBC, sumDC, aggKS, M-H [9], SVT [25], mlPCI [31], MSV [33], and WGC [39]. The introduction to these methods is as follows.

- 1) *sumPR, sumBC, and sumDC*: “sum” refers to the total sum of the values obtained by measuring the importance of nodes across different layers. We applied classical methods here, specifically PageRank (PR), betweenness centrality (BC), and degree centrality (DC).
- 2) *aggKS*: “agg” refers to the calculation of node importance values in an aggregated network. We applied the K-shell (KS) method for this purpose.
- 3) *M-H*: The M-H method represents the H-index of a node [8], which reflects the strength of connections between a node and its direct neighbors. All neighbors across all layers are taken into account in this method.
- 4) *SVT*: This method evaluates the importance of nodes in a multilayer network by combining hub and authority centrality vectors of both nodes and network layers.
- 5) *mlPCI*: This method generalizes the single-layer PCI (power community index) method to multilayer networks. By integrating multilayer features, it effectively addresses the challenge of identifying influential nodes.
- 6) *MSV*: The MSV method employs a localized hierarchical weighting approach to address the problem of node identification in multilayer networks.
- 7) *WGC*: The WGC method proposes a weighted gravity centrality approach for identifying important nodes in multilayer networks.

3) *Spreading Model*: To study the spreading process, we apply the susceptible-infected (SI) model [50], a classical model for disease propagation that is also widely used to simulate information spread in networks. In the SI model, individuals are classified into two types: susceptible (S) and infected (I).

The propagation mechanism of the SI model in multilayer networks is as follows: Each infected node i attempts to infect its neighboring node j in each layer with an infection probability λ . The propagation process operates independently in all layers, and once node i infects a neighboring node in a particular layer, the infection state will influence the propagation in other layers (i.e., the global infection state will be updated). Within each layer, edges propagate infection according to the specific infection probability of that layer. Propagation between different layers is independent, and cross-layer edges are not directly considered. However, the infection state of nodes is synchronized across all layers. The propagation effects across all layers can be regarded as a combined influence, with the experimental setup following an additive model.

At each time step in the simulation, the probability that a susceptible individual becomes infected by a neighboring infected node is denoted by λ . Since the model does not include recovery, nearly all individuals in a connected network will eventually become infected given enough time. It is important to note that we employed relatively small values of λ in our experiments to ensure that the total number of infected nodes does not increase too rapidly. Larger values of λ would result in rapid infection spread, complicating the analysis of the infection source, whereas smaller values allow for a more gradual spread, facilitating the assessment of individual node influence. Accordingly, we conducted experiments to identify which initial nodes, when activated, could trigger the maximum extent of infection.

4) *Evaluation Metrics*: We selected four standard evaluation metrics to assess the node identification performance across different methods.

1) Spearman Correlation Coefficient

Given two datasets, $X = [X_1, X_2, \dots, X_N]$ and $Y = [Y_1, Y_2, \dots, Y_N]$, the Spearman correlation coefficient [51] is calculated as follows:

$$\rho = 1 - \frac{6 \sum_{i=1}^N r_i^2}{N(N^2 - 1)} \quad (13)$$

where r_i represents the rank difference between X_i and Y_i , N is the total number of nodes in the network, and $\rho \in [-1, 1]$. A value of $\rho = -1$ indicates a perfect negative correlation, $\rho = 0$ denotes no correlation, and $\rho = 1$ signifies a perfect positive correlation. Higher values of ρ indicate stronger concordance between methods in terms of node identification effectiveness.

2) Imprecision Function

The imprecision function [52] is defined as

$$\epsilon(p) = 1 - \frac{M(p)}{M_{\text{eff}}(p)} \quad (14)$$

where $p \in [0, 1]$. $M(p)$ represents the average spreading influence of the top pN nodes with the highest importance, and $M_{\text{eff}}(p)$ denotes the average spreading efficiency of the top pN nodes with the highest spreading efficiency. This function quantifies the difference in the average spreading ability between the pN nodes ($0 < p < 1$) ranked highest in importance (based on a

given method) and the pN nodes with the highest spreading efficiency. A smaller ϵ value indicates greater accuracy in identifying influential nodes, with lower values reflecting a better match between the importance ranking and actual spreading efficiency.

3) Monotonicity Index

The monotonicity index, denoted as $M(R)$ [53], is defined as

$$M(R) = \left(1 - \frac{\sum_{r \in R} N_r(N_r - 1)}{N(N - 1)}\right)^2 \quad (15)$$

where R represents the ranking outcome, r denotes a specific ranking value within the result R , and N_r is the number of nodes assigned the ranking value r . A value of $M(R) = 1$ indicates that all nodes receive unique ranking values, demonstrating the maximum discriminative power of the ranking method. In contrast, a value of $M(R) = 0$ indicates that all nodes are assigned the same ranking value, representing the lowest possible discriminative power.

4) Network Connectivity

Network connectivity, as defined [54], is given by

$$Q = \frac{2 \sum_{i=1}^{N-1} \sum_{j=i+1}^N d_{ij}}{N(N - 1)} \quad (16)$$

where d_{ij} denotes the shortest path distance between node i and node j . A smaller value of Q signifies weaker connectivity within the network.

B. Experimental Results

1) Validity Experiments: In this experiment, Spearman correlation coefficients were used to assess the relationship between different node identification methods and node infection ability, as determined by the susceptible-infected (SI) model. To evaluate the correlation between these methods and actual infection ability, the node ranking results from each method and the node infection ranking over 10 time steps were treated as two distinct sequences. The correlation was calculated using (13). The infection ability of each node was represented by the number of other nodes it could infect when acting as the initial infected node over 10 time steps. Additionally, infection probability λ was varied between 0.02 and 0.2 for further testing. To ensure the reliability of the results, we averaged 100 independent simulation runs. The experimental outcomes are presented in Fig. 2.

In the Vickers network, our algorithm consistently outperforms all other methods. In the Kapferer network, it ranks highest, though it performs slightly below the sumDC and sumPR methods for $\lambda < 0.04$. For the Lazega network, our algorithm again demonstrates superior performance. Additionally, it yields optimal results in the CS-Aarhus network. In the London network, our algorithm not only significantly surpasses other methods but also exhibits a consistent upward trend. In the PierreAuger network, it corresponds to the highest curve, delivering the best overall performance. In both the Drosophila and Saccharomyces networks, our algorithm remains stable

while outperforming all other methods. Finally, in the Homo network, it achieves the highest performance across various λ values.

The previous experiment tested real-world networks, and our algorithm needs to maintain accuracy across diverse and large-scale networks. Therefore, we introduced a large-scale Barabási–Albert (BA) multilayer network with 10 000 nodes and used the Kendall correlation coefficient (τ) [54] to evaluate the algorithm's performance in handling large-scale networks. The Kendall coefficient is a statistical measure used to assess the consistency between two rankings. In this experiment, it quantifies the consistency between the node importance ranking and the infection capacity ranking. The Kendall correlation coefficient $\tau \in [-1, 1]$, where $\tau = 1$ indicates a highly consistent ranking and the best node recognition performance. This experiment first generates three BA scale-free networks, each containing 10 000 nodes, with each new node connected to six existing nodes (i.e., each new node is connected to six nodes in the existing network). Next, the three BA networks are combined into a multilayer network, where the node sets of each layer are identical, and nodes in the same spatial position across different layers are considered copies of the same entity. Finally, we quantify the consistency between the node importance ranking and the infection capacity ranking in the synthetic network. In large-scale networks, a smaller infection probability (λ) cannot effectively cover the entire network, affecting the reliability of the results. Therefore, we set the infection probability between 0.1 and 0.2. We averaged the results of 100 independent simulations to improve the reliability of the outcomes. The experimental results are shown in Fig. 3. The results indicate that the CWIIF algorithm significantly outperforms others in the node importance recognition task within multilayer networks and can effectively adapt to large-scale networks.

In addition, we compared the imprecision function values of different identification methods under the SI model. A smaller ϵ value indicates greater accuracy of the identification methods and superior node identification performance. This experiment aims to compare the propagation errors of the top pN nodes with the highest identification effectiveness and the top pN nodes with the greatest propagation influence. We first categorized the outcomes of each method to generate an ordered sequence of labels. Subsequently, we evaluated the infection capacity by analyzing the count of infected nodes at incremental 10-step intervals, thereby establishing a second sequence of labels. For our analysis, we focused on the initial 50% of both sequences ($p \in [0 - 0.5]$) and quantified recognition inaccuracies using (14). Each scenario was replicated 100 times to enhance result reliability, with averages depicted in Fig. 4.

Comparative analyses of network performance reveal distinct outcomes across different frameworks. Specifically, our algorithm excels in the Vickers network, surpassing other approaches. In the Kapferer network, while generally optimal, it performs slightly worse than the MSV and WGC methods when $p < 0.1$. Our algorithm achieves the best outcomes in the Lazega networks and remains competitive in the CS-Aarhus network, albeit slightly underperforming compared with the

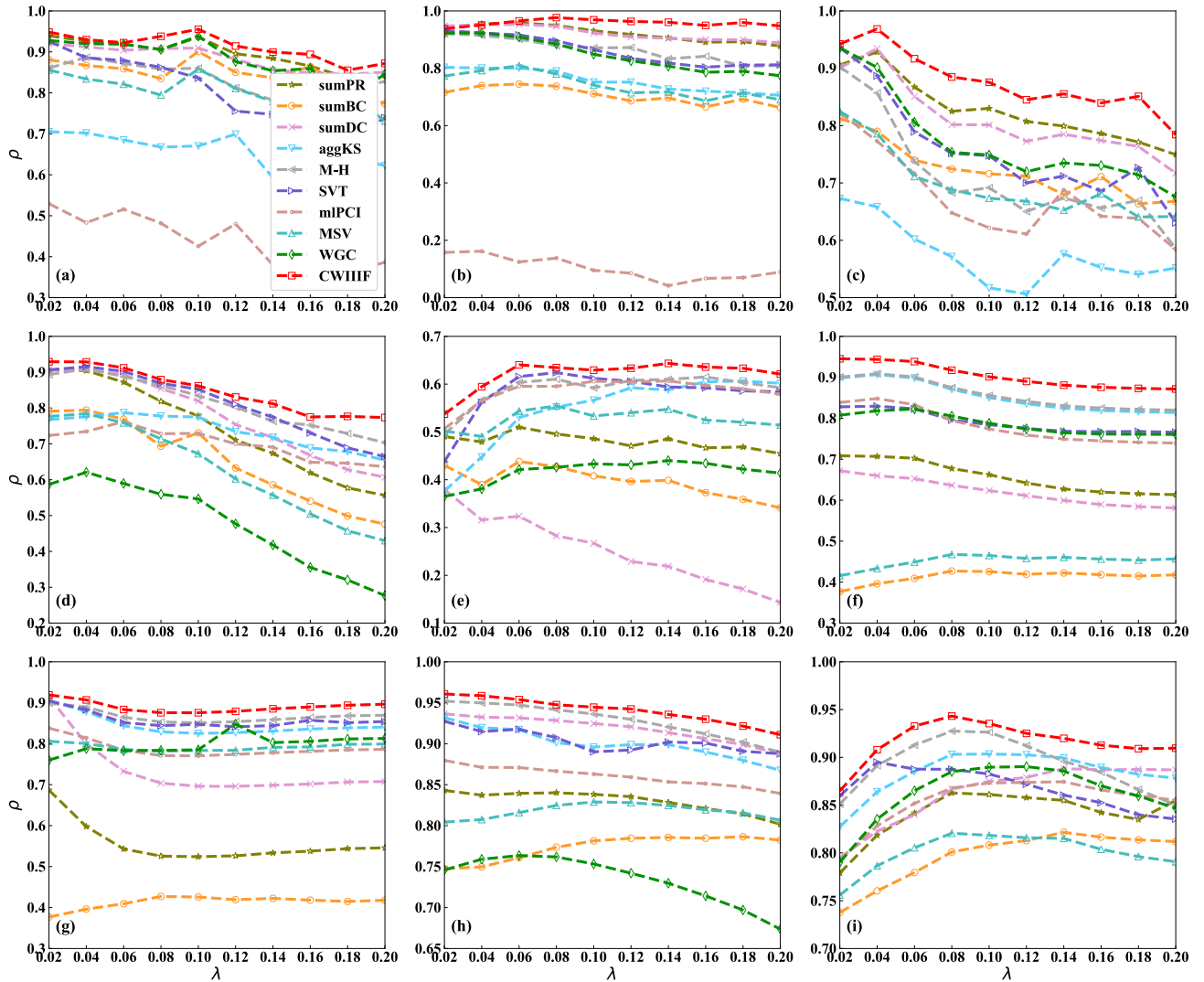


Fig. 2. Spearman correlation coefficient between 10 methods and the actual infection status across nine multilayer networks as λ varies. (a) Vickers-Chan-7thGraders_multiplex. (b) Kapferer-Taylor-Shop_multiplex. (c) Lazega-Law-Firm_multiplex. (d) CS-Aarhus_multiplex. (e) London_multiplex_transport. (f) PierreAuger_multiplex_coauthorship. (g) Drosophila_genetic_multiplex. (h) Sacchpomb_genetic_multiplex. (i) Homo_genetic_multiplex.

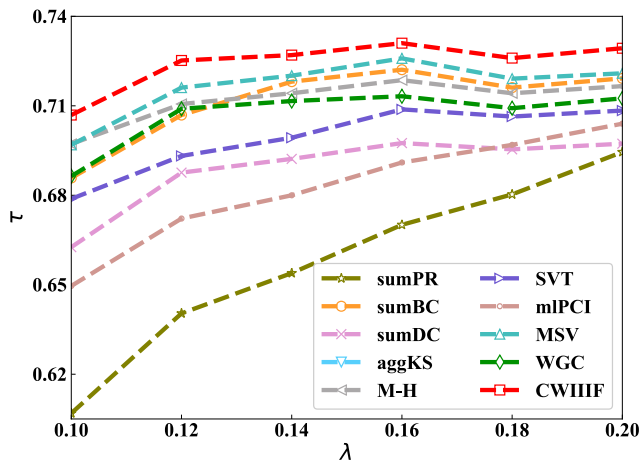


Fig. 3. Kendall correlation coefficient τ between the 10 methods and the actual node infection ability in a synthetic BA multilayer network with 10000 nodes, as λ varies.

mlPCI in the intermediate range. Despite minor limitations in the London network for $p < 0.1$ against the MSV, M-H, and WGC methods, our algorithm significantly outperforms others in overall efficacy. A similar pattern is observed in the PierreAuger network, where our algorithm closely competes with the aggKS method and excels beyond other competitors. In the Drosophila, Saccharomyces pombe, and Homo networks, our algorithm demonstrates superior performance and notable advantages.

2) *Monotonicity Experiments:* In general, robust node identification methods assign distinct values to uniquely identify each node. However, challenges emerge when multiple nodes exhibit identical influence scores, complicating their differentiation. To address this issue, it is crucial to monitor the occurrence of nodes sharing the same rank. As illustrated in Fig. 5, a high frequency of nodes with identical rankings indicates a reduced ability to differentiate between nodes. Compared with nine other methods, our algorithm demonstrates superior

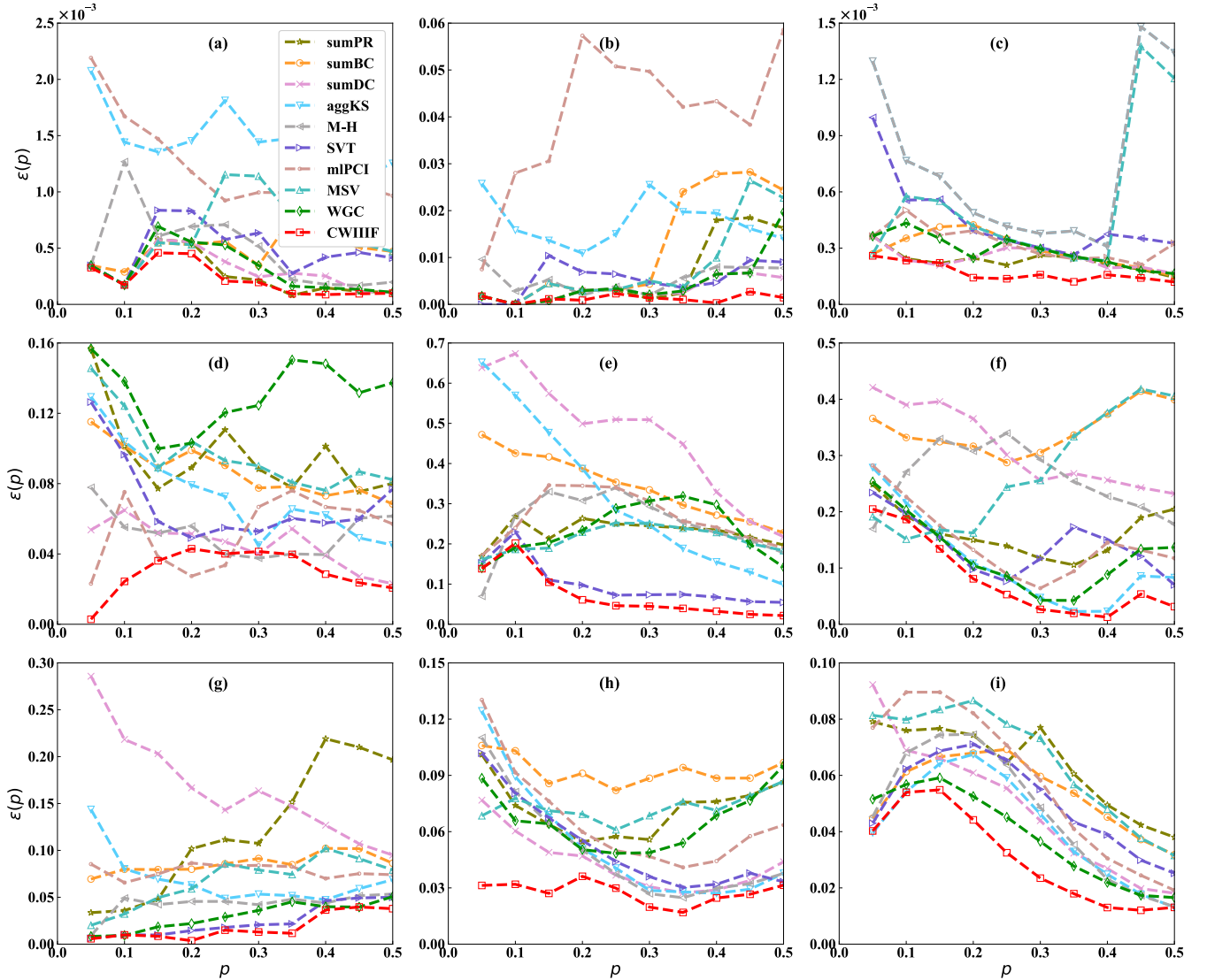


Fig. 4. Imprecision function values between ten methods and the actual infection situation in the nine multilayer networks when λ is 0.2. (a) Vickers-Chan-7thGraders_multiplex. (b) Kapferer-Taylor-Shop_multiplex. (c) Lazega-Law-Firm_multiplex. (d) CS-Aarhus_multiplex. (e) London_multiplex_transport. (f) PierreAuger_multiplex_coauthorship. (g) Drosophila_genetic_multiplex. (h) Sacchpomb_genetic_multiplex. (i) Homo_genetic_multiplex.

performance in distinguishing the influence of all nodes, particularly within the top 10% of influential nodes, where it shows significantly enhanced identification capabilities.

To objectively assess the identification capabilities of different methods, we utilized the monotonicity index and calculated monotonicity values using (15). As presented in Table II, with the leading method highlighted in bold, our algorithm consistently achieves monotonicity values closest to 1 across various networks. This result underscores our algorithm's superior ability to discriminate between nodes compared with other competing methods.

3) *Experiments on Resistance to Destruction:* In this study, we evaluated network robustness by selectively targeting and removing high-influence nodes, subsequently measuring the resulting connectivity. Node influence was initially ranked using various identification methods. We then removed the top $p \times 10\%$ of nodes based on their influence scores, along with

their associated edges. The remaining network connectivity after node removal was quantified using (16). To ensure the robustness of our results, we averaged the outcomes over 100 independent trials. The connectivity, represented by the Q value, is inversely related to network robustness: a lower Q value signifies diminished connectivity, thereby indicating the impact of removing high-influence nodes. The experimental results are shown in Fig. 6.

Our algorithm exhibited nuanced performance across different networks. In the Vickers network, it marginally outperformed competing methods. In the Kapferer network, it was the most effective overall. Comparisons in the Lazega network showed our algorithm closely competing with the aggKS method, yet it remained superior to other strategies. In the CS-Aarhus network, our method underperformed relative to mlPCI when $(0.3 < p < 0.6)$, but displayed the fastest decline in connectivity, indicating higher effectiveness. Similarly, in the

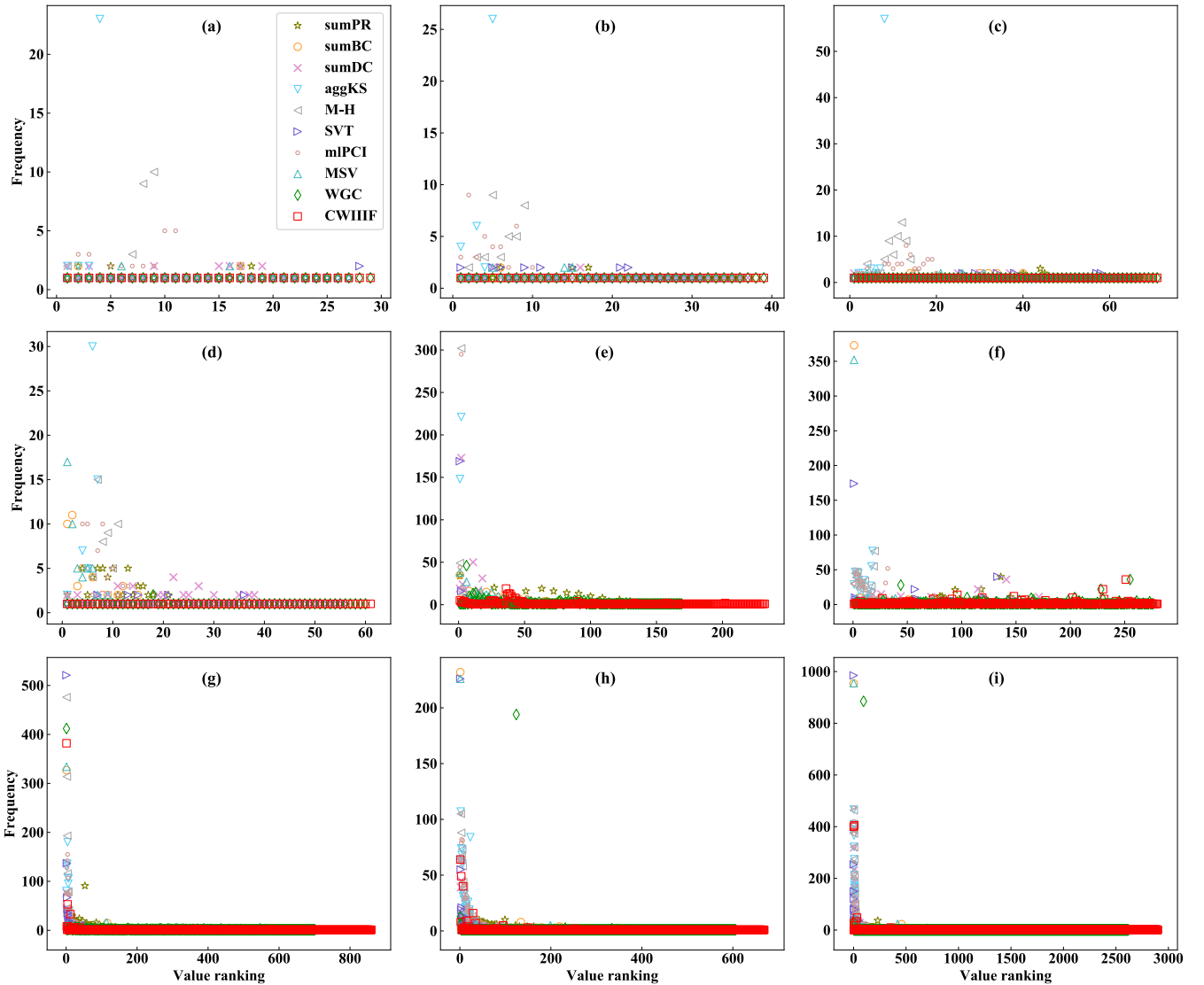


Fig. 5. Differentiated comparison of node importance rankings for 10 methods in the nine multilayer networks. (a) Vickers-Chan-7thGraders_multiplex. (b) Kapferer-Taylor-Shop_multiplex. (c) Lazega-Law-Firm_multiplex. (d) CS-Aarhus_multiplex. (e) London_multiplex_transport. (f) PierreAuger_multiplex_coauthorship. (g) Drosophila_genetic_multiplex. (h) Sacchpomb_genetic_multiplex. (i) Homo_genetic_multiplex.

TABLE II
MONOTONICITY VALUES OF 10 METHODS IN THE NINE REAL-WORLD MULTILAYER NETWORKS

Network	sumPR	sumBC	sumDC	aggKS	M-H	SVT	mlPCI	MSV	WGC	CWIIF
Vickers-Chan-7thGraders_multiplex	0.9951	0.9951	0.9658	0.1365	0.6251	0.9951	0.8531	0.9902	1	1
Kapferer-Taylor-Shop_multiplex	0.9919	0.9973	0.9946	0.2827	0.7624	0.9785	0.7933	0.9919	1	1
Lazega-Law-Firm_multiplex	0.9960	0.9952	0.9968	0.1254	0.8169	0.9952	0.9134	0.9960	1	1
CS-Aarhus_multiplex	0.9880	0.9761	0.9956	0.4726	0.7549	0.9945	0.8120	0.9902	0.9989	1
London_multiplex_Transport	0.9624	0.9706	0.5545	0.2321	0.0971	0.6212	0.1182	0.9706	0.9693	0.9826
PierreAuger_multiplex_Coauthorship	0.9738	0.2810	0.9749	0.8451	0.8474	0.7669	0.8833	0.2810	0.9753	0.9784
Drosophila_genetic_multiplex	0.8355	0.7521	0.7624	0.5852	0.6242	0.7025	0.6178	0.8034	0.8267	0.8453
Sacchpomb_genetic_multiplex	0.9588	0.7573	0.9569	0.7404	0.8876	0.8597	0.7730	0.7573	0.8915	0.9676
Homo_Genetic_multiplex	0.9557	0.8848	0.9246	0.8531	0.8655	0.8569	0.8477	0.8839	0.8810	0.9565

London network, while suboptimal for ($p < 0.2$), it generally outperformed others. Our algorithm achieved the best results in the PierreAuger network, despite a slight underperformance compared with the sumDC method between ($p < 0.2$). In the

Drosophila and Homo networks, our method was significantly superior, showing the fastest decline in connectivity. The results in the Sacchpomb network were comparable to others, yet overall, our algorithm was optimal.

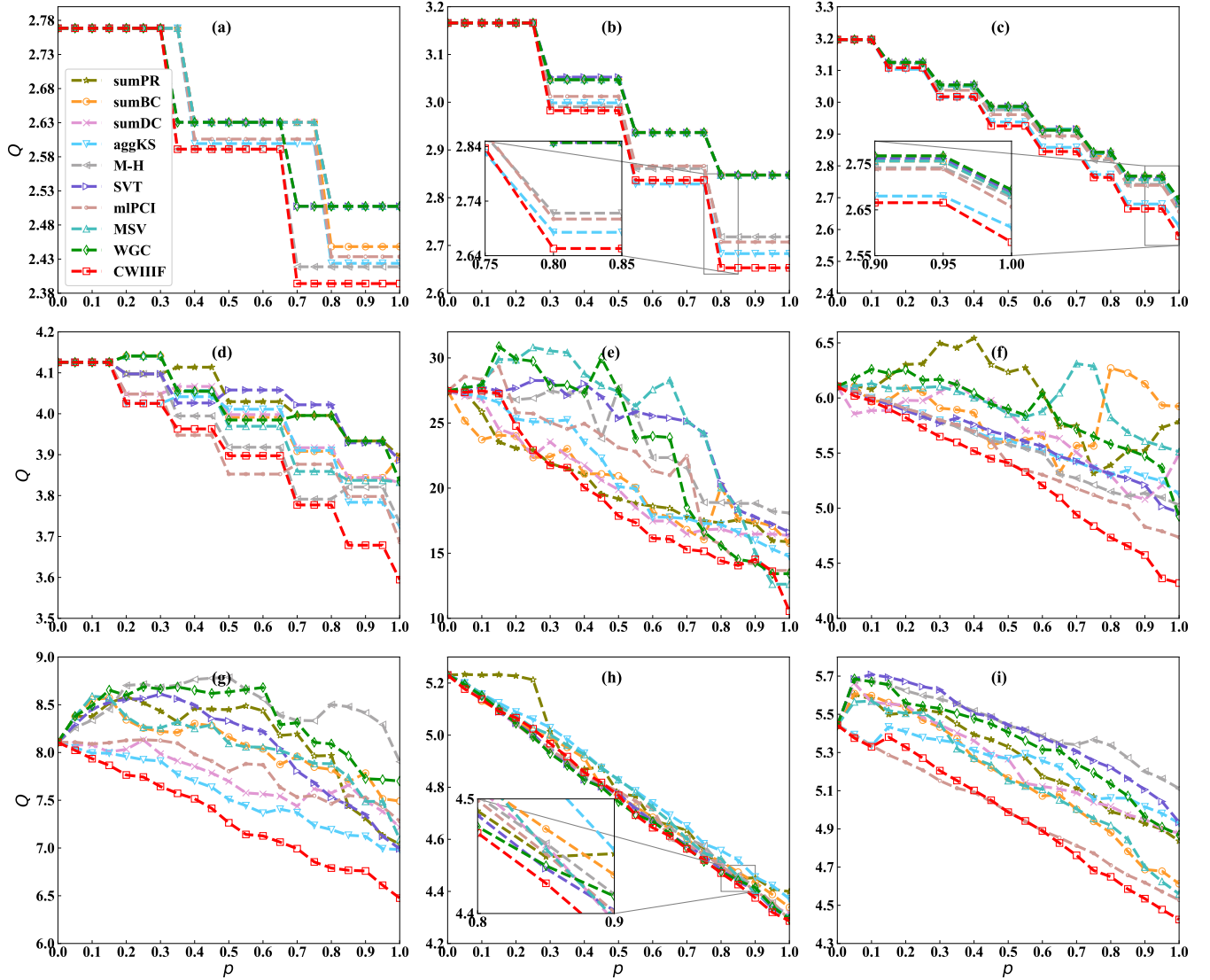


Fig. 6. Network connectivity sizes after the removal of $p \times 10\%$ of nodes across ten methods in nine multilayer networks. (a) Vickers-Chan-7thGraders_multilayer. (b) Kapferer-Taylor-Shop_multilayer. (c) Lazega-Law-Firm_multilayer. (d) CS-Aarhus_multilayer. (e) London_multilayer_transport. (f) PierreAuger_multilayer_coauthorship. (g) Drosophila_genetic_multilayer. (h) Sacchpomb_genetic_multilayer. (i) Homo_genetic_multilayer.

Based on the experimental results presented above, we conduct an in-depth analysis of the underlying reasons for the performance differences between our proposed algorithm and other methods. Our CWIIF algorithm not only considers the centrality of nodes within a single-layer network, but also captures both the global and local influences of nodes in multilayer networks through the fusion of hierarchical weighting and centrality vectors. In contrast, methods such as sumPR, sumBC, and sumDC only perform simple accumulations of single-layer centralities, neglecting the crucial inter-layer structural information, which leads to their relatively poor performance in multilayer networks. Unlike the aggregation-based aggKS method, our algorithm differentiates the importance of each layer and adapts to the complex structures of multilayer networks through weight allocation and the integration of inter-layer features. The aggKS method fails to reflect inter-layer differences, which

limits its effectiveness in networks with significant multilayer characteristics. The SVT method may be constrained by lower-order features in networks with complex structures, while our algorithm is more flexible in adapting to the structure of multilayer networks through weight distribution and inter-layer feature integration. Compared with methods such as M-H and WGC, which rely on direct or lower-order neighbor information and thus struggle to comprehensively assess node importance, our algorithm takes into account both the global and local influences of nodes across multilayer networks, allowing it to capture a more complete set of network information. The mlPCI method extends the single-layer PCI method to multilayer networks but fails to fully capture the structural differences between layers. The MSV method uses local hierarchical weighting but is limited by focusing on single-layer weighting, making it ineffective at handling inter-layer differences.

In contrast, our algorithm provides a more comprehensive and accurate evaluation of node importance by considering multilayer features.

IV. CONCLUSION

In this study, we tackle the challenge of identifying influential nodes within multilayer networks. We proposed a layered weighting approach that transforms each node's intra-layer influence vectors into scalar values, facilitating the calculation of influence scores and enabling the ranking of nodes by their influence within the network. The effectiveness and superiority of our proposed CWIIF algorithm were validated through four experiments on nine real-world networks, benchmarked against nine classical and state-of-the-art methods. Our validation process employed several metrics. First, we demonstrated a strong correlation between the TI values generated by our algorithm and the actual infection capabilities of nodes, as measured by the Spearman correlation coefficient and imprecision function within the SI model. Second, the CWIIF algorithm outperformed competing methods in identifying node importance, as evidenced by its superior performance on the monotonicity index. Finally, static attack simulations confirmed that the CWIIF algorithm identifies critical nodes more accurately than existing techniques. The experimental results highlight the robustness of the CWIIF algorithm in real-world network environments, establishing its superiority over current methodologies.

Future extensions of our work could proceed in three directions. First, we could further explore the concept of hierarchical weighting by incorporating additional and more relevant layer attributes that may influence network dynamics, thereby expanding the applicability of our approach. Moreover, integrating methodologies from single-layer network analysis could refine the calculation of intra-layer node importance. Second, given the current limitation of the CWIIF algorithm to unweighted and undirected networks, future research should extend its application to weighted and directed multilayer networks, thus improving its adaptability across various network analysis scenarios. Third, although our algorithm shows significant improvement in important node identification performance compared with existing advanced algorithms, it does not have an advantage in terms of runtime (time complexity). This limitation motivates us to further reduce its time complexity in future work.

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